Experimental proposal: divisibility onset, generator floor, and the backflow–divisibility dissociation in a passive $\mathcal{PT}$ photonic coupler

Jonathan Torkelson
Embeddetech, Inc., Tulsa, Oklahoma, USA `jtorkelson@embeddetech.com`

July 11, 2026

Summary. We propose a set of measurements on femtosecond-laser-written lossy waveguide couplers—the platform of Klauck *et al.* [2]—that would extract, for the first time, three process-level observables of the classical *transition-matrix process* of a $\mathcal{PT}$ dimer: the length t_c at which divisibility is first lost (permanently, in the broken phase), the asymptotic generator floor $-(a+\tilde{\kappa})$ (gain $a \equiv \gamma/2$, $\tilde{\kappa} \equiv \sqrt{a^2-\kappa^2}$), and the dissociation between information-flow Markovianity and divisibility, phase-locked to the $\mathcal{PT}$ transition. All three follow with no free parameters from the device constants; classical light suffices; and the primary observable is exactly invariant under the per-launch transmission-calibration errors, the leading multiplicative systematic (though not under the dark floor, which must be subtracted). A full measurement-chain simulation (`scripts/t2_measurement_chain.py`, public) underwrites the error budget quoted below.

Theory targets. Write the coupler as the dimer $H = \cos\theta\,\mathbf{1} + \mathrm{i}\sin\theta\,\sigma_z + s\,\sigma_x$; the passive device with coupling κ [cm^{-1}] and single-guide loss rate γ [cm^{-1}, amplitude convention, $\mathcal{PT}$-breaking at $\gamma = 2\kappa$ [2]] realizes $s \to \kappa$, $\sin\theta \to a \equiv \gamma/2$, $t \to z$ (propagation length) after removing the mean loss. The indivisible-process paper [1] establishes, for the measured intensity transfer matrix $\Gamma(z)$ (columns: launch guide; rows: output guide; index 1 the low-loss guide):

- *Divisibility onset:* P-divisibility first fails at t_c, with $t_c = \tilde{\kappa}^{-1}\,\mathrm{arcsinh}\big(\tilde{\kappa}/(\kappa\sqrt{2})\big)$, $\tilde{\kappa} = \sqrt{a^2-\kappa^2}$ (continued to arcsin in the unbroken phase), exactly where $\det\Gamma$ crosses zero; in the broken phase ($\gamma > 2\kappa$) the resulting indivisibility is *permanent* (P-divisible precisely on $[0, t_c)$), in the unbroken phase it recurs in windows of period $\pi/\sqrt{\kappa^2-a^2}$, and t_c is continuous through the exceptional point.
- *Generator floor:* in the broken phase the off-diagonal generator entry satisfies $L_{12}(z) < -(a+\tilde{\kappa})$ for all $z > t_c$ and *rises* to the strictly negative constant $-(a+\tilde{\kappa})$ as $z \to \infty$: a floor on $|L_{12}|$, approached from below by the signed entry.
- *Dissociation:* on the same data, the survival-conditioned total-variation distance between the two launch columns shows *negligible* backflow in the late window (for onsets t_c preceding it) of the broken phase while divisibility fails terminally—numerically, the two standard notions of non-Markovianity come apart, phase-locked to the $\mathcal{PT}$ transition.

Why this is new. The apparatus exists and is characterized as a function of length [2]; exceptional-point criticality of information flow has been measured [4]; and indivisibility without information backflow has already been observed on one tomographic data set—Bernardes *et al.*'s "weak" non-Markovian regime, together with the transition to the strong one [5]. What no experiment has extracted, to our knowledge, is a closed-form divisibility-onset *length* or a generator *floor*, nor phase-locked the dissociation to a $\mathcal{PT}$ transition on a passive photonic coupler. Note that Klauck *et al.* operated well inside the unbroken phase (far from the exceptional point at $\gamma/2\kappa = 1$); the broken phase—where the terminal behavior lives—is unexplored territory for these observables.

Devices and protocol. Fabricate four couplers (D1–D4) spanning three $\mathcal{PT}$ regimes (all at $\lambda = 815\,\mathrm{nm}$ in fused silica; loss by sinusoidal modulation [3]; per sample, one Hermitian reference coupler, one straight reference guide, and one straight modulated-loss guide—the latter carrying the device modulation in isolation, so its length-resolved transmission pins the loss rate γ independently of the coupling—on the same chip):

device	κ [cm^{-1}]	γ [cm^{-1}]	regime	predicted t_c [cm]	predicted floor [cm^{-1}] (D3: window period)
D1	0.10	0.60	broken ($\gamma/2\kappa = 3$)	5.10	-0.583
D2	0.1125	0.90	broken, deeper ($\gamma/2\kappa = 4$)	3.98	-0.886
D3	0.50	0.30	unbroken	1.55 (first window)	period 6.59 cm
D4	0.30	0.60	*at* the EP ($\gamma/2\kappa = 1$)	2.36 ($= 1/\kappa\sqrt{2}$)	-0.300 ($= -a$ asymptote, $\tilde{\kappa} = 0$)

For each family, a ladder of chip lengths $z = 0.5, 1.0, \ldots, 10\,\mathrm{cm}$ (20 samples; Klauck-style varying-length sets). At each length, launch classical laser light into each guide in turn and record both output intensities relative to the straight reference guide: four numbers per length assemble $\hat{\Gamma}(z)$. The required loss rates (2.6–7.8 dB/cm power loss) are set by the tunable-bending-loss modulation technique [3], which reports radiation losses of almost arbitrary amount; we note that the highest loss demonstrated on this same platform in [2] is $\approx$3 dB/cm, so D2's requirement sits a factor $\sim$2.6 above the published same-platform record. D2, the deepest working device, keeps its whole measurement—onset and floor—within 6 cm chips. A single-photon version (heralded pairs, as in [2]) would reproduce the numbers below at 10^4 detected photons per launch; entangled-pair extensions [6, 7] address the conditioning-signal observable treated elsewhere in this program and are out of scope here.

Estimators and error budget (simulated). The measurement-chain simulation models per-sample fabrication disorder (2% on κ, a), per-launch transmission-calibration error (3%), photon shot noise, and a 10^{-4} dark/crosstalk floor, with 400 Monte Carlo realizations per configuration:

- t_c *is calibration-invariant by construction:* the onset is estimated from the zero crossing of det of the *column-normalized* transfer matrix, which is exactly invariant under per-launch intensity rescalings—an algebraic identity, and the simulated calibration-only spread is correspondingly zero. Precision at 10^4 photons/launch: ± 0.19 cm (D1), ± 0.28 cm (D2); at 10^6 with all systematics, ± 0.09 and ± 0.06 cm (Klauck's HOM-dip benchmark: 0.05 cm [2]). The dark floor is the one systematic this estimator does *not* absorb: it biases the crossing by $+0.05$ cm (D1) and $+0.12$ cm (D2), and must be subtracted. The bias grows with $\gamma/2\kappa$ because the surviving diagonal at t_c weakens; by $\gamma/2\kappa \approx 5$–6 the determinant no longer cleanly crosses zero at this dark level and the estimator degrades beyond usefulness, its bias and spread exceeding the signal, which is what sets D2's depth.
- *The floor requires the global fit:* pointwise finite-difference generators are fragile where the survival probability approaches the dark floor (documented in the script's null control). Fitting the closed-form model to the full ladder, with per-column calibration nuisances, recovers the floor to $\pm 3.4\%$ (D1) and $\pm 4.1\%$ (D2) for both broken devices—29σ (D1) and 24σ (D2) below zero—already at 10^4 photons/launch. This is the significance of the floor *parameter* under the closed-form model, not a model-independent test of permanence.
- *Dissociation:* on identical data and analysis, the late-window ($z \geq 5$ cm) conditional backflow is 0.018 (D2, whose onset $t_c = 3.98$ cm precedes the window—the clean broken-phase instance, dropping to 0.004 at 10^6 photons) and 0.015 (D1, whose $t_c = 5.10$ cm sits at the window edge, so this retains the theory-predicted onset-neighborhood tail—≈ 0.01 noiseless, 0.009 as

simulated at 10^6, the dark floor slightly suppressing it—rather than pure noise) versus 0.49 for D3 (unbroken), while the fitted floor is $\geq 24\sigma$ negative for D1/D2: the dissociation is resolved with more than an order of magnitude of margin.

- *Unbroken control:* D3's first window position is recovered to $\pm 0.03\,\mathrm{cm}$ and the window period to $\pm 2.3\%$, confirming the recurrent (non-terminal) signature. (The exceptional-point device D4 is not in the simulated budget; continuity of t_c through the EP is an analytic property of the closed form, not a simulated result here.)
- *Two dimensionless points:* $\kappa\, t_c$ and floor$/\kappa$ depend on $\gamma/2\kappa$ alone, so D1 and D2 test the closed form at $\gamma/2\kappa = 3$ and 4 ($\kappa\, t_c = 0.510$ and 0.447) rather than re-testing one prediction at two overall scales.

What a positive result establishes. These would be the first process-level (transition-matrix) observables of $\mathcal{PT}$ dynamics: they test the identification of broken-$\mathcal{PT}$ evolution with survival-conditioned dynamics of a dilated stochastic process—in particular that the broken phase is *backflow-Markovian at late lengths yet permanently indivisible*, a prediction with no analogue in the quantum-state-based analyses of [4, 8]. A null result—most directly, a genuine *restoration* of divisibility ($\det \hat{\Gamma}$ crossing back positive) in a resolvable window just past t_c, a model-independent signal; or a systematic late-window lack of fit of the closed form (e.g. L_{12} relaxing toward zero once its asymptote is freed from (a, κ) in an extended fit)—would falsify the closed forms and, with the device Hamiltonian independently constrained by the on-chip reference structures, the dilation reading for this system. The restoration test must be read only where the divisor determinant is resolvable: since $\det \hat{\Gamma} \to 0^-$ as $z \to \infty$ (survival, and with it every intensity, vanishing), a naive late-z sign reading is powerless—at the 10^{-4} dark level and 10^4 photons per launch the measured late-ladder determinant flips positive in 79% of realizations even when the closed forms hold exactly (falling to 9% at 10^6)—so a genuine restoration must clear the measured det noise-plus-dark envelope in the window just past t_c, not merely change sign at large z. The lack-of-fit falsifier is informative already at 10^4 detected photons per launch (an afternoon of data on this platform); the model-independent restoration test additionally requires photons enough to resolve $\det \hat{\Gamma}$ above the dark floor across that window.

Provenance. Prepared with the large language model Claude Fable 5 (Anthropic), directed by the author; simulation and error budget reproducible from `scripts/t2_measurement_chain.py` in the public repository https://github.com/embeddetech/p02-measurable-ghosts (also archived in the program hub, https://github.com/embeddetech/ghosts-program).

References

[1] J. Torkelson, *Indivisible stochastic processes for non-Hermitian and higher-derivative quantum systems: detailed balance, broken $\mathcal{PT}$ symmetry, and the explosion of ghosts*, paper 1 of this program, repository https://github.com/embeddetech/p01-indivisible-ghosts (2026).

[2] F. Klauck *et al.*, *Observation of $\mathcal{PT}$-symmetric quantum interference*, Nat. Photonics **13**, 883 (2019), arXiv:1904.08135.

[3] T. Eichelkraut *et al.*, *Radiation-loss management in modulated waveguides*, Opt. Lett. **39**, 6831 (2014).

[4] L. Xiao *et al.*, *Observation of critical phenomena in parity-time-symmetric quantum dynamics*, Phys. Rev. Lett. **123**, 230401 (2019).

[5] N. K. Bernardes *et al.*, *Experimental observation of weak non-Markovianity*, Sci. Rep. **5**, 17520 (2015).

[6] J.-S. Tang *et al.*, *Experimental investigation of the no-signalling principle in parity–time symmetric theory using an open quantum system*, Nat. Photonics **10**, 642 (2016).

[7] M. Ehrhardt, M. Heinrich, and A. Szameit, *Observation-dependent suppression and enhancement of two-photon coincidences by tailored losses*, Nat. Photonics **16**, 191 (2022).

[8] K. Kawabata, Y. Ashida, and M. Ueda, *Information retrieval and criticality in parity-time-symmetric systems*, Phys. Rev. Lett. **119**, 190401 (2017).